

SAT-SA — What We Built and Why

1. Problem Statement

SUPERVISORY BOTTLENECK

NCIIPC supervises Critical Sector Entities (CSEs). Manual supervisory reviews of SOC evidence have repeatedly uncovered weaknesses not visible through policies, audits, self-assessments, management reports, KPI dashboards, or compliance documentation. **However, manual review does not scale effectively across hundreds of CSEs.**

2. Our Solution

SAT-SA TOOL

SAT-SA is a supervisory analytics tool that analyzes raw SOC evidence to automatically identify **execution gaps, negative space, behavioral anomalies, and peer deviations** for targeted supervisory action.

3. Core Paradigm Shift

TRADITIONAL SECURITY SYSTEMS	SAT-SA SUPERVISORY SYSTEM
Asks: "Is there a cyberattack?"	Asks: "Is the SOC actually functioning effectively?"
Focus: Threat Detection / Telemetry	Focus: SOC Audit & Evidence Quality

✓ Current Project Status

MVP VERIFIED

337 / 337 Unit & Integration Tests Passing	100% Seeded Weakness Detection
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5. Presentation Roadmap

- **Problem & Approach:** Execution gaps vs negative space
- **Technical Architecture:** Fast-API, SQLite, Analytics Pipeline
- **Signal Catalog:** 22 Signals across 5 taxonomy categories
- **Validation Results:** Empirical results on 50 CSE dataset
- **Live Demo Script:** 2-minute step-by-step rehearsal guide
- **Remaining Work & Q&A:** Technical debt, planned signals, Q&A

Understanding the Problem

TRADITIONAL METRICS ASK

"Did the security process appear to happen?"

MANUAL SUPERVISORY REVIEW ASKS

"Did the process actually work as designed?"

Execution Gaps

OPERATIONAL GAMING

- **Shallow Investigation:** Alerts acknowledged but closed without substantive notes.
- **Velocity Gaming:** Cases closed unnaturally quickly to satisfy MTTR KPIs.
- **Unescalated Criticals:** High-severity alerts marked benign without escalation rationale.
- **Template Automation:** Identical copy-paste investigation descriptions across cases.
- **Unmonitored Controls:** Security tools deployed but telemetry unreviewed.
- **Metric Gaming:** Behaviors optimized to look good on executive dashboards.

Negative Space

ABSENT TELEMETRY

- **Missing Telemetry:** Critical assets or server groups producing zero logs.
- **Category Absences:** Expected alert types (e.g., auth failures) completely missing.
- **Missing Escalations:** Zero escalated incidents despite high incident volumes.
- **Unexpected Quiet:** Sudden unnatural drops in baseline alert ingestion.
- **Blind Spots:** Sub-networks completely absent from security coverage.

Key Supervisory Insight: ABSENCE OF EXPECTED EVIDENCE IS ITSELF AN EVIDENCE-BACKED SIGNAL.

8 CORE CAPABILITY AREAS

1. THREAT DETECTION
2. INVESTIGATION
3. ESCALATION
4. INCIDENT RESPONSE
5. SECOPS
6. GOVERNANCE
7. OPERATIONAL DISCIPLINE
8. CYBER RESILIENCE

EXPLICITLY OUT OF SCOPE

SAT-SA is **NOT** a SOC, **NOT** a SIEM, **NOT** real-time monitoring, **NOT** a national surveillance platform, and **NOT** a continuous log collector.

How We Think About the Problem

P1: Supervisory, Not Ops

Others build threat detection for SOC analysts asking *"Is there an attack?"*

We build **SOC audit analytics for NCIIPC examiners** answering *"Is the SOC functioning effectively?"*

P2: Evidence, Not AI

Relies primarily on deterministic analytics (z-scores, change-point detection, peer benchmarks).

Finding → Signal → Metric → Records

LLM is optional & for explanation only. No black-box scores.

P3: Negative Space Focus

Standard anomaly detection looks for *unusual occurrences*.

SAT-SA flags when **expected evidence is completely absent** (e.g., CSE-089 has 217 endpoints but 0 endpoint alerts).

Hybrid Analytics Architecture

Deterministic core with modular qualitative extensions

Layer 1: Statistical Engine

VALIDATED

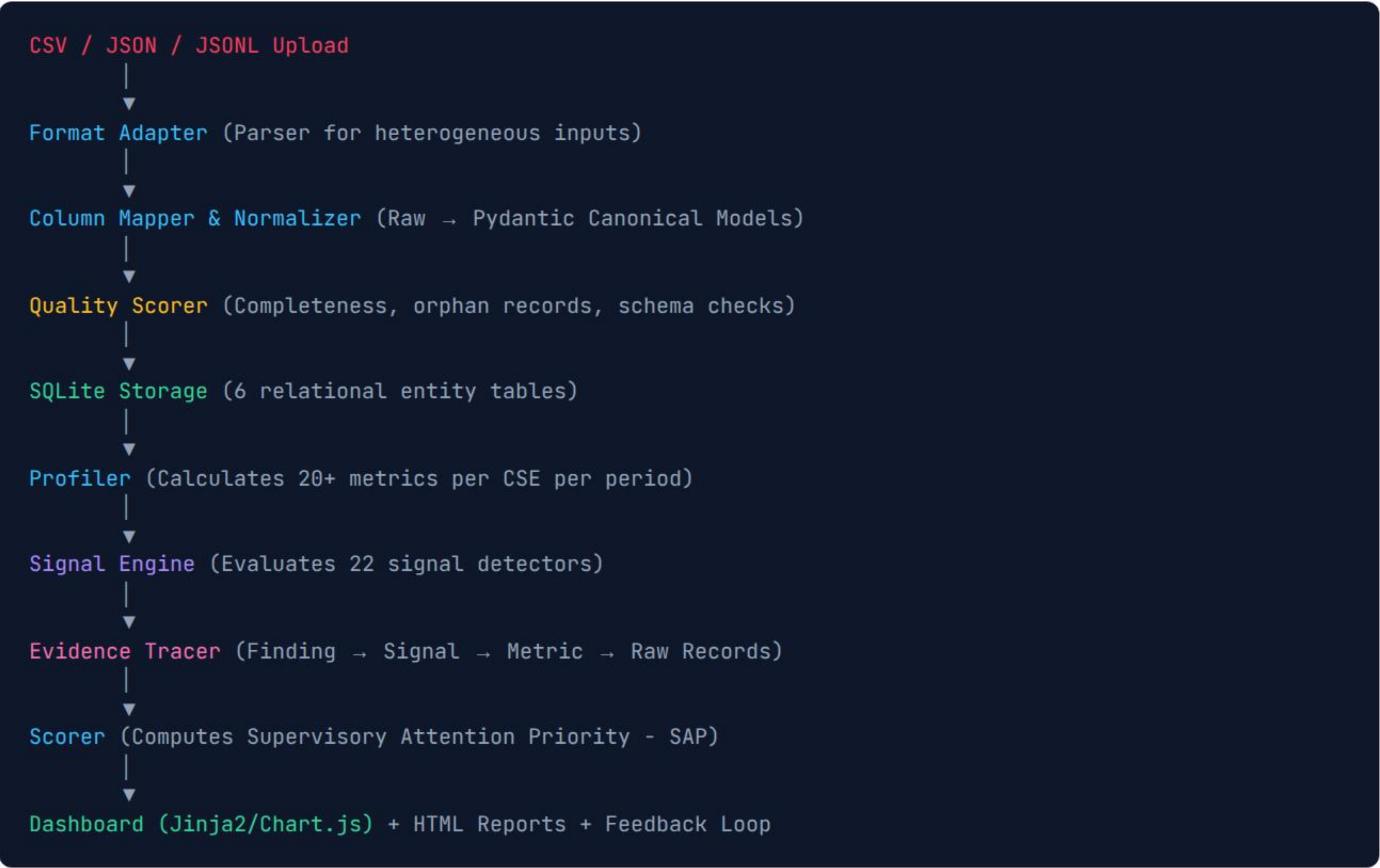
- Timing metrics (MTTR, velocity variance)
- Volume analysis & telemetry baselining
- Cross-CSE peer group z-score benchmarking

Layer 2: Reasoning Quality Engine

PLANNED

- Investigation note depth parsing (heuristics)
- Semantic keyword & template matching
- Escalation rationale validation

DATA PROCESSING PIPELINE



TECH STACK SPECIFICATION

Component	Technology	Rationale
Language	Python 3.9+	Offline-first, standard libraries
Backend	FastAPI	Auto OpenAPI, high performance
Database	SQLite	Zero-config, air-gap ready
Analytics	Pandas / SciPy	Statistical routines, no ML overhead
Frontend	Jinja2 / JS	No Node.js build step required
LLM (Opt)	Ollama	Local inference for offline labs

Key Architectural Decisions

- Optional Fields:** Missing attributes become `None` rather than raising fatal exceptions.
- Adapter Pattern:** Extensible parser decouples analytics logic from input schema formats.
- Envelope Response:** API returns uniform `{data, meta, errors}` wrapper.
- Advisory Feedback:** User feedback recorded for review, never auto-mutates rules.

22 Signals Across 5 Categories

1. Execution Gaps

5/5 IMPLEMENTED

- `superficial_closure`
Fast close + thin notes
- `escalation_without_action`
Escalated but ignored
- `quality_degradation`
Change-point drop
- `severity_mismatch`
High-sev marked benign
- `kpi_divergence`
Metrics vs quality gap

2. Negative Space

5/5 IMPLEMENTED

- `alert_volume_gap`
Expected vs actual volume
- `missing_investigations`
High-sev with zero notes
- `missing_alert_categories`
Absent alert types
- `telemetry_absence`
Critical assets silent
- `escalation_absence`
Zero escalations made

3. Behavioral Anomalies

5/5 IMPLEMENTED

- `temporal_drift`
Sudden depth change
- `unusual_quiet_period`
No alerts in active hrs
- `bulk_closure_pattern`
Mass closures on days
- `shift_variance`
Quality delta across shifts
- `recurring_incident`
Repeated un-resolved alert

4. Peer Deviation

3/3 IMPLEMENTED

- `closure_velocity_outlier`
Z-score vs peer group
- `investigation_depth_outlier`
Z-score vs peer group
- `escalation_rate_outlier`
Z-score vs peer group

5. Reasoning Quality

0/4 IMPLEMENTED

- `shallow_justification`
High-sev minimal notes
- `template_notes`
Identical texts across cases
- `missing_escalation_rationale`
Escalated with no reason
- `reasoning_inflation`
Unevidenced actions claimed

Proving It Works

VALIDATION PERFORMANCE (50 CSE DATASET)

METRIC	VALIDATION RESULT	TARGET	STATUS
Coverage	100% (8/8)	≥ 70%	EXCEEDED
Precision	100%	≥ 60%	EXCEEDED
False-Positive Rate	0%	< 40%	EXCEEDED
Examiner Alignment	0.909	≥ 0.70	EXCEEDED
Pipeline Speed	90s / 50 CSEs	< 30 min	EXCEEDED

Test Suite Coverage

337 tests passing across 18 test files (Schemas, Ingestion, Profiler, Evidence, Benchmarking, Scoring, Signals, API, Dashboard, Feedback).

SEEDED WEAKNESS DETECTION VERIFICATION

TARGET CSE	INJECTED WEAKNESS	DETECTED SIGNAL
CSE-042	Quality Drop / Drift	quality_degradation, temporal_drift
CSE-017	Shallow Review	superficial_closure
CSE-089	Missing Telemetry	missing_alert_categories
CSE-031	Uninvestigated Sev1	missing_investigations
CSE-055	Rapid Close Outlier	closure_velocity_outlier
CSE-073	Zero Escalations	escalation_absence
CSE-019	Copy-paste Notes	template_investigation
CSE-061	Peer Depth Outlier	investigation_depth_outlier

Honest Scope Disclaimer: These metrics reflect performance on our controlled 50-CSE synthetic validation dataset (8 seeded + 42 clean). They do not imply universal production accuracy.

2-Minute Demo Script

```
$ python scripts/demo.py → http://localhost:8000/dashboard/
```

PRE-CONFIGURED DEMO MODE

TIMESTAMP	SCREEN / ACTION	TALKING POINTS / SCRIPT
0:00 - 0:10	Portfolio Overview	"50 CSEs analyzed. Key metrics look green on top-level dashboards, but SAT-SA identifies underlying supervisory anomalies that traditional KPIs miss."
0:10 - 0:30	Click CSE-042 (Rank #1)	"8 findings aggregated into a supervisory case. Notice investigation depth declined by 69% across 4 quarters while ticket closure velocity increased."
0:30 - 0:50	Finding Detail & Evidence	"Every finding links directly to raw evidence records. Statistical change-point occurred in Q3 — prompting examiners to ask what operational shift occurred."
0:50 - 1:10	Navigate to CSE-089	"217 endpoints deployed, yet zero endpoint alerts generated. Demonstrates negative space detection — auditing the expected evidence that is absent."
1:10 - 1:30	Peer Benchmarking	"CSE-042 displays a z-score of -1.15 against its peer cohort. This is not a marginal discrepancy, but a statistically significant operational deviation."
1:30 - 1:50	Examiner Feedback Loop	"Examiner flags finding as worthwhile. Feedback is recorded into audit log to refine future scoring without unverified auto-mutation."
1:50 - 2:00	Download Executive Report	"Generates evidence-backed, auditable PDF reports. SAT-SA identifies potential weaknesses; the human examiner makes the final decision."

What's Left and Who Does What

REMAINING TASK MATRIX

TASK DESCRIPTION	EFFORT	OWNER	STATUS
Reasoning Quality Layer (4 Signals)	2 Days	TBD	PLANNED
Synthetic Investigation Notes	1 Day	TBD	PLANNED
Demo Video Backup Recording	1 Day	TBD	PLANNED
Final Slide Deck Polish	1 Day	TBD	IN REVIEW
Team Rehearsal Run-through	0.5 Day	All	PENDING

Key Files for Code Walkthrough

src/analytics/schemas.py
src/evidence/tracer.py

src/analytics/signal_engine.py
scripts/run_pipeline.py

ANTICIPATED Q&A PREP CHEAT-SHEET

Q: How does SAT-SA differ from a traditional SIEM?

"A SIEM monitors logs to detect threat actors. SAT-SA audits SOC evidence to verify operational effectiveness."

Q: What is the primary technical novelty?

"Systematic negative space detection combined with transparent evidence-traced supervisory scoring."

Q: Why not rely on standard KPI dashboards?

"KPIs like MTTR can be easily gamed by fast, superficial closures. SAT-SA explicitly detects metric gaming."

Q: How does SAT-SA handle missing or sparse fields?

"Missing data becomes None values and triggers quality warnings, preventing crashes or skewed scoring."